Thirteen Integer Variables through a Single Five-Radical Norm

Explicit consistency candidates, ordinary proofs, and a precise calibration boundary

Research contribution prepared for
Vladimir Reshetnikov's ProveIt project

24 October 2026

Repository snapshot: `9a33342a03d19cf5b061b2083c6b5d3f27f752ed`

Abstract

We give a completely specified polynomial in thirteen unrestricted integer variables, of exact total degree 1088, evaluated by a division-free circuit with 195 live binary arithmetic operations. A simpler companion has degree 1392 and 186 operations. Both use the six fixed coefficients already present in ProveIt's source-34/source-71 construction. There are no free parameters, variable exponents, hidden existential coordinates, or new coefficient-majorant payloads.

The new mechanism places a guard inside a radical norm. Four radicands are positive before any Diophantine condition is known. A zero of the five-radical norm therefore forces its remaining, guard radicand to be nonnegative. That fact supplies strict positivity inequalities; those inequalities then make the five radical magnitudes strictly superincreasing. Rationality and integrality follow, restoring four eliminated square witnesses and the main Pell congruence. An integer translation lowers the degree from 1408 to 1392. A second thirteen-variable construction uses one carrier for two coprime congruences to reduce the exact degree further to 1088, with 195 binary operations. We retain the degree-1392 formula as the simpler base presentation and prove the additional CRT variant separately.

The algebraic reduction is proved here. Its interpretation as a consistency equation composes explicitly identified, inherited ordinary native-compiler and K215 calibration results. At that ordinary theorem level, ZFC proves that the polynomial has no integer zero if and only if $\mathrm{Con}(\mathrm{ZFC})$. This contribution does not supply a new kernel-checked calibration theorem or the repository's still-open encoded object-ZFC derivation. It proves neither consistency nor global minimality. The archive also supplies lower-degree companions, coefficient reconstruction, exact norm and degree audits, and editable sources.

Contents

1 Results and proof status **3**
1.1 Three distinct proof levels . . . 3

2 The fixed polynomial **4**

3 A division-free norm circuit **5**

4 The guard establishes its own hypotheses **5**

5 Automatic separation of five radicals **6**

6 Recovering all integer witnesses **7**
6.1 Restoration of the original strong kernel . . . 8

7 Canonical completeness and consistency calibration **9**

8 Exact degrees of the five-root equations **10**
8.1 The unshifted degree 1408 . . . 10
8.2 The translated degree 1392 . . . 11

9 A coprime carrier: thirteen integers at degree 1088 **11**
9.1 Complete specification . . . 11
9.2 Unconditional realness, including the zero-scale branch . . . 12
9.3 Separation with only $|Y| \geq |q|$ . . . 13
9.4 Two coprime congruences, with no hidden scale promise . . . 14
9.5 Exact degree and a coefficient certificate . . . 15

10 Four-root and three-root companions **16**
10.1 Fourteen integers, degree 696 . . . 17
10.2 Fifteen integers, degree 344 . . . 18

11 Four quotient-fusion companions **18**

12 Reproduction and evidence **19**

13 What is new and what remains to improve **20**

A Complete bounded coefficient recipe **20**

B Archive map and integration **22**

1 Results and proof status

The request for a simplest consistency equation has several competing objectives: unrestricted integer unknowns, total degree, circuit size, coefficient size, and completeness of the proof certificate. We report explicit tradeoffs. Every variable in Table 1 ranges over $\mathbb{Z}$.

Construction	Unknowns	Exact degree	Binary operations
Five-root norm with coprime carrier	13	1088	195
Five-root norm, translated	13	1392	186
Five-root norm, unshifted	13	1408	185
Four-root norm, centered	14	696	150
Four-root norm, unshifted	14	792	137
Three-root norm, centered	15	344	131
Quotient fusion	20	24	120
Quotient fusion and transport	19	28	116
Quotient fusion and transport	18	32	112
Quotient fusion and transport	17	88	110

Table 1: New literal equations. Operations are live binary $+, -, \times$ gates with shared identical subexpressions; fixed numeral leaves are not gates. This does not measure bit complexity or expanded monomial count.

The inspected repository's earlier thirteen-integer construction has degree 10 461 744. On that specific variable/degree comparison, 1392 reduces the degree by a factor exceeding 7500, and 1088 by a factor exceeding 9600. This is not a variable-count record: incoming material already gives a ten-integer construction of degree 1 245 324. Nine *positive* variables constitute a different domain comparison [1, 2, 3]. Sun's recent ten-unknown theorem is also relevant [10].

The formal product of conjugate radicals is a classical relation-combining technique. The new feature is the absorption of an indeterminate-sign guard without a circular proof or large separating weights. The determinant guard supplies its own quantitative bound.

1.1 Three distinct proof levels

1. **New algebraic theorem.** The thirteen-variable polynomial is existentially equivalent to the explicit normalized system in Section 6. Its denominator argument, square extraction, and exact degree are proved here.
2. **Inherited ordinary calibration.** The source-34 compiler, signed/residue-oriented native theorem, canonical Pell construction, and ordinary K215 theorem identify the relevant witnesses with finite syntactic ZFC refutations. We use these named ordinary results as imported mathematical lemmas.
3. **Formal certificate status.** Finite checks establish identities, coefficients, circuits, and degree certificates. They do not prove an unbounded compiler theorem. The repository separately records that the complete native calibration and actual encoded `BProv ZFCax_s [] (fIff ...)` endpoint remain open formalization obligations [4, 5].

Thus a claim of ZFC provability below means ordinary mathematical provability by composition with the specified ordinary proofs. It does not mean that a completed Lean/Rocq or encoded object-language derivation is delivered.

2 The fixed polynomial

The thirteen unknowns, in literal circuit order, are

$$\boxed{j,\ C,\ z,\ h,\ r,\ v,\ w,\ T,\ c,\ n,\ t,\ p_2,\ p_3 \in \mathbb{Z}.}$$

All other letters are polynomial abbreviations; C and c are different unknowns. Use the complete `mixed-parity-unrounded-1064921` tuple

$$\beta = 2^{1064921} - 1, \qquad K_0, \quad M_C, \quad M_F,$$

with the *already shifted* $M_F = M_{F,\text{native}} + \beta$, and

$$A_{\text{cone}} = 2^{1113}, \qquad B_{\text{cone}} = 2^{666104}.$$

Appendix A specifies the finite coefficient recipe. The archive contains all six exact numeral payloads, the complete tile and machine data, and an independent reconstruction. No coefficient is selected by a halting search or a consistency assumption.

Define the following registers:

$$\begin{aligned} J &= 6j+1, & q &= \beta J + 1, & W &= 12w + 4, \\ X &= qW, & Y &= (2v+1)q^3, & E &= XY, \\ a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1. \end{aligned} \tag{2.1}$$

The transport equation is built into

$$\begin{aligned} F &= (K_0 + W)C - z(q-1), & \alpha &= q - F - C, \\ R &= \big(q(q-F) - C\big)(q^2-1) + (M_C + qM_F)J, & & \\ k &= R + 1 + hE, & \eta &= c - Yk, & \zeta &= k - \eta. \end{aligned} \tag{2.2}$$

The abbreviation α is only used in the proof. Set

$$\begin{aligned} I &= (2r+1)c^2, & D_f &= 1 + \Delta I^2, & Q &= \Delta^2 I^2, \\ \delta &= Q - 1, & V &= D_f(cT - R) - c, & C_y &= QV^2 - 1, \\ D_\tau &= 1 + XY^2(XY^2+1)k^2, & D_{\text{m}} &= 1 + \Delta c^2, & b_0 &= X + ac. \end{aligned} \tag{2.3}$$

Finally define

$$\begin{aligned} U &= A_{\text{cone}} F, & Z_c &= B_{\text{cone}} C, \\ G &= UZ_c - (q - U - Z_c)^2, & P_\eta &= \eta\zeta, \\ A_{\text{g}} &= GP_\eta - (t^2 - G - P_\eta + 1)^2 - 1 - p_2^2 - p_3(p_3+1). \end{aligned} \tag{2.4}$$

Let the universal five-root polynomial be

$$\mathcal{F}_{32}(x; A_0, \ldots, A_4) = \prod_{\varepsilon \in \{-1,1\}^5} \left(x - \sum_{i=0}^{4} \varepsilon_i \sqrt{A_i} \right). \tag{2.5}$$

Changing any formal radical's sign permutes the factors, so the product belongs to $\mathbb{Z}[x, A_0, \ldots, A_4]$. Each radicand has degree at most 16. Our equation is the cleared polynomial

$$\boxed{P_{13} = \delta^{16} \mathcal{F}_{32}\left(H(n+V) + b_0;\ D_{\text{m}}, H^2 D_f, H^2 D_\tau, H^2 A_{\text{g}}, \frac{H^2 C_y}{\delta} \right) = 0.} \tag{2.6}$$

The denominator is canceled *symbolically*. No division or square-root operation occurs in the actual equation. Section 3 gives a recurrence using only integer polynomial arithmetic, and `equations/norm13_degree1392.json` supplies every circuit gate.

Replacing $n + V$ by n gives the degree-1408, 185-operation companion. The two zero sets are related by the invertible integral translation $n_{\text{old}} = n_{\text{new}} + V$; V does not involve n.

3 A division-free norm circuit

The three-root norm is

$$\mathcal{F}_8(x; A, B, C) = \big((x^2 - A - B - C)^2 - 4(AB + AC + BC)\big)^2 - 64ABCx^2.$$

Its cleared version is given by

$$\begin{aligned} J_N &= \delta(x^2 - A - B) - C_{\mathrm{n}}, \\ T_N &= \delta^2 AB, \qquad & R_N &= \delta C_{\mathrm{n}}, \\ K_N &= J_N^2 - 4\big(T_N + R_N(A + B)\big), \\ N_3(x; A, B, C_{\mathrm{n}}, \delta) &= K_N^2 - 64 T_N R_N x^2. \end{aligned} \tag{3.1}$$

This is $\delta^4 \mathcal{F}_8(x; A, B, C_{\mathrm{n}}/\delta)$ after clearing.

To obtain the five-root norm, calculate $N_3(x - u - v)$ in the free rank-four algebra with basis $1, u, v, uv$, $u^2 = D$, $v^2 = B_g$. Squaring (a_0, a_1, a_2, a_3) gives

$$\begin{aligned} b_0 &= a_0^2 + D a_1^2 + B_g a_2^2 + D B_g a_3^2, \\ b_1 &= 2(a_0 a_1 + B_g a_2 a_3), \qquad & b_2 &= 2(a_0 a_2 + D a_1 a_3), \\ b_3 &= 2(a_0 a_3 + a_1 a_2). \end{aligned} \tag{3.2}$$

The square of $x - u - v$ is $(x^2 + D + B_g, -2x, -2x, 2)$. If the resulting N_3 has vector (f_0, f_1, f_2, f_3), set

$$\begin{aligned} g_0 &= f_0^2 + B_g f_2^2 - D(f_1^2 + B_g f_3^2), \\ g_1 &= 2(f_0 f_2 - D f_1 f_3), \qquad & N_5 &= g_0^2 - B_g g_1^2. \end{aligned} \tag{3.3}$$

Exactly,

$$N_5 = \delta^{16} \mathcal{F}_{32}(x; A, B, D, B_g, C_{\mathrm{n}}/\delta).$$

For (2.6) use $A = H^2 D_f$, $B = H^2 D_\tau$, $D = D_{\mathrm{m}}$, $B_g = H^2 A_{\mathrm{g}}$, $C_{\mathrm{n}} = H^2 C_y$. The identity holds with zero, negative, or dependent radicands; the algebra need not be a field of degree 32. The independent audit constructs the conjugate product in reverse elimination order and compares all 20 349 nonzero coefficients, including the cleared dictionary.

4 The guard establishes its own hypotheses

Lemma 4.1 (Unconditional realness)**.** *For every integer input, $\delta \neq 0$, $H \neq 0$, and*

$$D_{\mathrm{m}} > 0, \quad D_f > 0, \quad D_\tau > 0, \quad D_y := C_y/\delta \geq 1.$$

Proof. The fixed β is odd, $J = 6j + 1 \neq 0$, and $|q| \geq \beta - 1 \geq 16$. Thus q is even. Since $W = 4(3w + 1) \neq 0$, $|X| \geq 64$, $|Y| \geq 4096$, and $|a| \geq (|X| - 1)|Y| > 18$. Consequently

$\Delta = (a+2)^2 - 1 > 1$. Also a is even, so Δ and H are odd. The first two radicands are positive, and $D_\tau \geq 1$ because $u(u+1) \geq 0$ for the integer $u = XY^2$.

If $c = 0$, then $Q = 0, \delta = -1, C_y = -1, D_y = 1$. If $c \neq 0$, then $Q > 1$ and $D_f = 1 + \Delta(2r+1)^2 c^4 > |c| + 1$. For $b = cT - R$, $b = 0$ gives $|V| = |c| \geq 1$; otherwise $|V| \geq D_f - |c| > 1$. Hence

$$D_y = V^2 + \frac{V^2 - 1}{Q - 1} \geq 1.$$

Both cases exclude $\delta = 0$. □

Lemma 4.2 (Joint positivity)**.** *Put* $\mathrm{Pair}(x, y, t) = xy - (t^2 - x - y + 1)^2$. *For integers* x, y, *some* t *has* $\mathrm{Pair}(x, y, t) > 0$ *if and only if* $x, y > 0$. *Every positive integer is a value of*

$$\mathrm{Pos}(p_1, p_2, p_3) = 1 + p_1^2 + p_2^2 + p_3(p_3 + 1).$$

Proof. Positivity of Pair gives $xy > 0$. If $x = -a, y = -b$ with $a, b > 0$, then $(t^2 + a + b + 1)^2 > ab$, a contradiction. Conversely let $N = x + y - 1 \geq 1$; $xy \geq N$. For $m = \lfloor \sqrt{N} \rfloor$, choose $t = m$ if $N - m^2 \leq m$, and $t = m + 1$ otherwise. Then $|t^2 - N| \leq m$. The error is zero when N is square, and otherwise its square is less than $N \leq xy$.

For $s > 0$, $4s - 3$ is 1 or 5 mod 8, so Legendre's three-square theorem applies. Modulo 4, exactly one square is odd. Writing the squares as $(2p_1)^2, (2p_2)^2, (2p_3 + 1)^2$ gives the formula for s. This is the inherited joint-guard lemma [6]. □

Proposition 4.3 (Realness before squarehood)**.** *At every integer zero of* P_{13},

$$A_{\mathrm{g}} \geq 0, \qquad G > 0, \qquad \eta\zeta > 0, \qquad c \neq 0, \quad Q > 1.$$

Proof. Lemma 4.1 permits interpretation of the cleared norm over $\mathbb{C}$. A linear factor vanishes. Four radicals and the center are real. If $A_{\mathrm{g}} < 0$, the fifth radical $H\sqrt{A_{\mathrm{g}}}$ is the only nonzero imaginary term, which is impossible. Thus $A_{\mathrm{g}} \geq 0$. Now

$$\mathrm{Pair}(G, \eta\zeta, t) = A_{\mathrm{g}} + 1 + p_2^2 + p_3(p_3 + 1) > 0.$$

Lemma 4.2 gives both strict guards, without first assuming that A_{g} is square. If $c = 0$, then $\eta\zeta = -Y(Y+1)k^2 \leq 0$, a contradiction. □

This order avoids a circular argument: realness establishes the guards; the guards will establish separation; separation will establish squarehood.

5 Automatic separation of five radicals

All bounds here precede Pell-index or tableau decoding. Put

$$x = |X| \geq 64, \quad m = |Y| \geq 4096, \quad K = |k|, \quad C_0 = |c|.$$

Since η, ζ are nonzero integers of the same sign, $K \geq 2$, $0 < \theta = \eta/k < 1$, and $c = k(Y + \theta)$. Thus

$$K(m-1) < C_0 < K(m+1). \tag{5.1}$$

Write $f_+ = \sqrt{D_f}$, $\tau_+ = \sqrt{D_\tau}$, $y_+ = \sqrt{D_y}$, $d_+ = \sqrt{D_{\mathrm{m}}}$, $g_+ = \sqrt{A_{\mathrm{g}}}$. The last root may be zero.

For $a_* = |a| \geq (x-1)m$, $\sqrt{\Delta} > a_* - 3 > xm/2$. Hence

$$f_+ > \frac{xm}{2} C_0^2, \qquad f_+ > 3C_0^2, \qquad K(xm^2 - 1) < \tau_+ < K(xm^2 + 1).$$

The numerical lower bounds imply

$$\tau_+ > 2C_0, \quad f_+ > 2\tau_+, \quad Q > D_f > D_\tau > 1, \quad Q > C_0^2. \tag{5.2}$$

For instance $f_+ > 2\tau_+$ follows from $mK(m-1)^2 > 4(m^2+1/x)$. For $b = cT - R$, two cases give

$$\begin{aligned} b = 0: &\quad C_0 < y_+ < C_0 + 1, \\ b \neq 0: &\quad |V| \geq f_+^2 - C_0 > 2f_+, \quad y_+ > |V| > 2f_+. \end{aligned} \tag{5.3}$$

In the first case the correction $(C_0^2 - 1)/(Q-1)$ is less than 1.

The main root satisfies

$$\frac{C_0}{5} < \frac{d_+}{|H|} < \frac{C_0}{2}. \tag{5.4}$$

Use $\sqrt{\Delta} > a_* - 3$, $|H| \leq 4a_* + 3$ for the lower bound, and $d_+ < (a_*+3)C_0$, $|H| \geq 4a_* - 3$ for the upper. The determinant identity

$$12G = 4q^2 - \big(3(U + Z_c) - 4q\big)^2 - 3(U - Z_c)^2 \tag{5.5}$$

gives $0 < G \leq q^2/3 < q^2$. Since $\eta\zeta \leq k^2/4$, the positive subtracted term in (2.4) implies

$$g_+ < \frac{|q|K}{2} < \frac{C_0}{10}. \tag{5.6}$$

The last step uses $m \geq |q|^3$ and (5.1). The huge determinant coefficients disappear from this estimate.

Proposition 5.1 (Strict separation)**.** *After omitting a zero g_+, each entry in either order below exceeds the sum of all earlier entries:*

$$\begin{aligned} b = 0: &\quad g_+,\ d_+/|H|,\ y_+,\ \tau_+,\ f_+, \\ b \neq 0: &\quad g_+,\ d_+/|H|,\ \tau_+,\ f_+,\ y_+. \end{aligned}$$

Proof. The two small roots have sum less than $0.6C_0$, with $d_+/|H| > 2g_+$. For $b = 0$,

$$y_+ + d_+/|H| + g_+ < 1.6C_0 + 1 < 2C_0 < \tau_+,$$

and $y_+ > C_0$, $f_+ > 2\tau_+$. For $b \neq 0$, successively use $\tau_+ > 2C_0$, $f_+ > 2\tau_+$, $y_+ > 2f_+$. □

6 Recovering all integer witnesses

Lemma 6.1 (Square classes)**.** *If positive numbers with rational squares have strictly superincreasing magnitudes, and a signed sum is rational, then each is rational.*

Proof. In the multiquadratic extension they generate, a nonrational root is changed in sign by some Galois automorphism. Subtract its image of the rational sum from that sum. The resulting nonempty signed subsum cannot vanish: its largest magnitude exceeds the sum of all others. Independence of the original square classes is not assumed. □

Lemma 6.2 (Adjacent-coefficient descent)**.** *If $Q > 1$, V are integers and $y \in \mathbb{Q}$, then*

$$QV^2 - (Q-1)y^2 = 1 \quad \Longrightarrow \quad y \in \mathbb{Z}.$$

Proof. Take absolute values and write $y = a/d$ in lowest terms. The equation implies $d^2 \mid Q-1$. At $|V| = 1$, $|y| = 1$. For $V, y > 0$, $V > 1$, use

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV.$$

The norm is unchanged, V' is integral, and the reduced denominator of y' remains d, since $2Q-1 \equiv 1 \bmod d$. The inequalities $y > V$ and $(Q-1)y < QV$ give $-V < V' < V$. The norm excludes $V' = 0$. Descent reaches $|V| = 1$ without changing the denominator. □

Definition 6.3 (Normalized integer system)**.** Let $\mathcal{K}$ use the same registers, omit n, and add the five integer coordinates f, τ, y, γ, p_1, with equations

$$\begin{aligned} f^2 &= D_f, & \tau^2 &= D_\tau, & & \\ QV^2 - (Q-1)y^2 &= 1, & (b_0 + H\gamma)^2 &= D_{\mathrm{m}}, & A_{\mathrm{g}} &= p_1^2. \end{aligned} \tag{6.1}$$

This is a seventeen-integer system, with no rational notation in its defining equations.

Theorem 6.4 (New algebraic reduction)**.** *P_{13} has an integer zero if and only if $\mathcal{K}$ has an integer solution. This assertion and proof are formalizable in ZFC.*

Proof. At a zero, Proposition 4.3 gives the guards, then Proposition 5.1 gives separation. Divide the vanishing factor by H. Lemma 6.1 makes its five roots rational. The signed roots d, f, τ, p_1 of the integer radicands are integers; Lemma 6.2 makes the signed auxiliary root y integral. Absorb the factor's signs into these roots:

$$H(n+V) + b_0 = d + H(f + \tau + p_1 + y).$$

Therefore

$$\gamma = n + V - f - \tau - p_1 - y = (d - b_0)/H \in \mathbb{Z}. \tag{6.2}$$

This restores (6.1). Conversely, given its witnesses, set

$$n = \gamma + f + \tau + p_1 + y - V. \tag{6.3}$$

The guard gives $c \neq 0, \delta > 0$, and the specified signed roots make one factor zero. Every quantity is chosen before n; there is no circular witness selection. □

6.1 Restoration of the original strong kernel

Let $i = 2r + 1$, and define

$$\begin{aligned} I_{\mathrm{old}} &= \Delta i c^2, & r_{\mathrm{old}} &= (\Delta i - 1)/2, \\ o_{\mathrm{old}} &= f(cT - R), & j_{\mathrm{old}} &= D_f T - 1 - R\Delta i^2 c^3. \end{aligned} \tag{6.4}$$

The first quotient is integral because Δi is odd. Using $f^2 = D_f$, polynomial identities give

$$\begin{aligned} I_{\mathrm{old}} &= (2r_{\mathrm{old}} + 1)c^2, & I_{\mathrm{old}}^2 &= \Delta(f^2 - 1) = Q, \\ fo_{\mathrm{old}} - c &= V, & cj_{\mathrm{old}} - R &= V. \end{aligned} \tag{6.5}$$

This restores both old quotients without an inferred divisibility shortcut. The first centered parent root is $\tau - XY^2k$; the main root is $b_0 + H\gamma$. The transport row vanishes identically:

$$(K_0 + W)(q - F - \alpha) - F - z(q-1) = 0.$$

The old radix coordinate is $J_3 = 2j$, and $A_{\mathrm{g}} = p_1^2$ restores exactly the joint guard with $\mathrm{Pos}(p_1, p_2, p_3)$. These are the incoming signed, residue-oriented, ordinary-strong equations [6, 7]. The new normalization chooses a subfamily of strong witnesses. Completeness uses the canonical regeneration proved next, rather than claiming that every old witness is already normalized.

7 Canonical completeness and consistency calibration

The preceding restoration proves soundness into the inherited signed kernel. Completeness requires a new canonical choice of the strong and auxiliary witnesses. It does not require every old solution to admit this normalization with its original auxiliaries.

Write the ordinary Pell sequences as

$$\chi_A(s) + \psi_A(s)\sqrt{A^2-1} = (A+\sqrt{A^2-1})^s, \qquad s \in \mathbb{N}.$$

The inherited canonical native witness has $A = a+2 > 1$ even, $R \equiv 3 \pmod 4$, and

$$c = \psi_A(R) > 0, \qquad D = \chi_A(R).$$

The coefficient proof gives $q = 2^e$ with e odd, odd Y/q^3, and the residue-oriented scale $X/(4q) \equiv 1 \pmod 3$. Also $J = (q-1)/\beta$ is odd and 1 mod 3, so $J = 6j+1$. Thus the stronger radix parity in (2.1) preserves canonical existence [9, 7].

Lemma 7.1 (Odd normalized strong witness)**.** *Put*

$$I_* = \psi_A(Rc), \qquad f = \chi_A(Rc), \qquad i = I_*/c^2.$$

Then i is a positive odd integer and $f^2 = 1 + \Delta i^2 c^4$.

Proof. The composition identity gives $I_* = c\psi_D(c)$. Expanding $(D + c\sqrt{\Delta})^c$, the coefficient of $\sqrt{\Delta}$ is

$$I_* = \sum_{\substack{1 \le s \le c \\ s \text{ odd}}} \binom{c}{s} D^{c-s} c^s \Delta^{(s-1)/2}.$$

The $s = 1$ term is $c^2 D^{c-1}$, and every other term has a factor c^3. Hence $c^2 \mid I_*$. Since A is even, the recurrence $\psi_A(s+2) = 2A\psi_A(s+1) - \psi_A(s)$ gives $\psi_A(s) \equiv s \pmod 2$. Consequently c and I_* are odd, and division by the odd integer c^2 makes i odd. This parity conclusion does not rely on the parity of the leading binomial term. The Pell identity gives the displayed norm. □

Set $r = (i-1)/2$ and $S = \Delta I_*$. The inherited squared-modulus auxiliary lemma [8] gives

$$V_* = \frac{\chi_S(R)}{S} \in \mathbb{Z}_{>0}, \qquad y = \psi_S(R) > 0,$$

with

$$S^2 V_*^2 - (S^2-1)y^2 = 1, \qquad V_* \equiv -c \pmod{f^2}, \qquad V_* \equiv -R \pmod c.$$

Its hypotheses are $R \equiv 3 \pmod 4$, $c = \psi_A(R)$, $f^2 - \Delta I_*^2 = 1$, and $c \mid I_*$, all established above. Since $f^2 \equiv 1 \pmod c$, $\gcd(c, f) = 1$, and therefore

$$T = \frac{V_* + c + Rf^2}{cf^2} \in \mathbb{Z}.$$

It gives $V_* = f^2(cT - R) - c$, exactly (2.3). The main native coordinates and their determinant/strip guards remain unchanged. Choose their joint positivity representation and then choose n by (6.3). This completes the canonical reverse construction.

Theorem 7.2 (Ordinary consistency equivalence)**.** *Use the fixed tuple in Appendix A and the pinned ordinary native, signed-domain, and K215 calibration theorems. Then*

$$\mathrm{ZFC} \vdash \left[\left(\forall \mathbf{x} \in \mathbb{Z}^{13}\ P_{13}(\mathbf{x}) \neq 0 \right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}) \right].$$

The same assertion holds for each of the companion polynomials below, with its stated integer dimension.

Proof. The new argument restores an inherited source witness from every zero. The canonical construction above produces a zero from every canonical native witness. The ordinary source theorem identifies existence of native witnesses with blank halting of the fixed K215 table, and the ordinary K215 theorem identifies that halting statement with existence of a finite syntactic ZFC refutation [9, 5]. These arguments use ordinary finite coding, induction, Pell arithmetic, first-order completeness, and the inherited set-theoretic calibration, all as mathematics in ZFC. Compose the equivalences and negate them. □

The dependencies in this theorem are explicit imported mathematical results. In particular, no inference of internal provability is made merely from the external Lean statement

```
BlankHalts machine215 ⟷ BProv ZFCax_s [] Form.fBot.
```

The ordinary K215 exposition itself states a ZFC-provable equivalence. Its external Lean proof passes through the numerical calculus, ZF models, and constructible-choice calibration. The repository's machine-checked translation to its actual sealed `Form` consistency sentence remains open [4]. The archive's `provenance/CALIBRATION_STATUS.md` separates these claims. If the imported ordinary calibration is treated as an obligation to be re-established, the new unconditional contribution is the exact source-zero equivalence; the final consistency interpretation then depends on that calibration. Neither bounded testing nor proof-assistant soundness is substituted for the missing proof artifact.

8 Exact degrees of the five-root equations

Every fixed numeral has degree zero, and each supplied unknown has degree one. In the base coordinates,

register	H	Δ	R	D_f	Q	V	C_y	D_τ	A_g
degree	6	12	5	18	30	23	76	34	44.

These are degrees after the literal substitutions, not degrees on the solution set.

8.1 The unshifted degree 1408

The four integer radicands in (2.6) have degrees $14, 30, 46, 56$; the auxiliary numerator H^2C_y has degree 88, while $\deg \delta = 30$. The universal norm is weighted homogeneous of weight 32, with weight one for the center and weight two for each radicand. In the unshifted equation the center has degree seven. A term using the auxiliary radicand to power s has degree at most

$$30(16-s) + 88s + 56(16-s) = 1376 + 2s \le 1408.$$

Only $s = 16$ reaches 1408, and its coefficient is $+1$.

To show that this term survives actual coefficient specialization, put

$$\begin{aligned} q_0 &= 6\beta j, & W_0 &= 12w, & Y_0 &= 2vq_0^3, & X_0 &= q_0W_0, \\ a_0 &= X_0Y_0, & i_0 &= 2r, & F_2 &= W_0C - q_0z, & R_0 &= -q_0^3F_2. \end{aligned} \tag{8.1}$$

Then

$$(H^2C_y)_{88} = 16a_0^{10}i_0^6c^{12}R_0^2,$$

a nonzero polynomial for the authentic fixed $\beta > 0$. The unshifted degree-1408 part is its sixteenth power.

8.2 The translated degree 1392

For the circuit of Section 3, use

$$D = D_{\mathrm{m}}, \quad B_g = H^2 A_{\mathrm{g}}, \quad A = H^2 D_f, \quad B = H^2 D_\tau, \quad C_{\mathrm{n}} = H^2 C_y.$$

Give the two formal root symbols u, v weights $7, 28$. A coefficient on $1, u, v, uv$ then has its ordinary degree plus $0, 7, 28, 35$. Multiplication respects this filtration.

For $s = Hn + b_0$, the center is $x = s + HV$. The scalar part of J_N, evaluated at $x + u + v$, has the exact cancellation

$$(J_N)_0 = \delta(s^2 + 2HVs + D + B_g - A - B) - H^2(V^2 - 1).$$

Its degree is 86, led by δB_g. The four weighted component degrees are

$$86, \quad 66, \quad 87, \quad 65.$$

The unique largest component is $2Q_0 x_0 v$, of weight 87. Its square has leading scalar $(2Q_0 x_0)^2 (B_g)_0$, of degree 174; the correction in K_N has degree at most 164. The correction in N_3 has weight at most 312, so N_3 has unique leading scalar

$$16 Q_0^4 x_0^4 (B_g)_0^2$$

of degree 348. Each of its four conjugates has this same leading scalar. Their product therefore has exact degree $4 \cdot 348 = 1392$.

Explicitly,

$$\eta_0 = -hX_0 Y_0^2, \quad Q_0 = a_0^4 i_0^2 c^4, \quad V_0 = -a_0^2 i_0^2 c^4 R_0, \quad x_0 = 4 a_0 V_0, \quad (B_g)_0 = -16 a_0^2 \eta_0^4.$$

Thus

$$(P_{13})_{1392} = 2^{80} a_0^{128} i_0^{64} c^{128} R_0^{16} \eta_0^{32} \neq 0.$$

For a directly checkable specialized certificate,

$$\left[j^{800} w^{160} v^{192} h^{32} r^{64} c^{128} z^{16} \right] P_{13} = 2^{1456} 3^{960} \beta^{800} > 0.$$

This proves exact degree, rather than relying on cancellation-free gate estimates or modular interpolation.

9 A coprime carrier: thirteen integers at degree 1088

The preceding construction can be reduced further without adding a variable or a fixed coefficient. The explicit factor q^3 in the scale $Y = (2v + 1)q^3$ is expensive after repeated substitution. We instead supply $Y = qv$, then recover the missing divisibility and odd quotient through the same root norm that recovers the main Pell congruence. The two relevant moduli are provably coprime on every integer assignment. This gives an explicit polynomial of exact degree 1088 in the same thirteen unrestricted integer unknowns.

9.1 Complete specification

The coordinates remain

$$j, C, z, h, r, v, w, T, c, n, t, p_2, p_3 \in \mathbb{Z}.$$

Use the same six fixed numerals $\beta, K_0, M_C, M_F, A_{\mathrm{cone}}, B_{\mathrm{cone}}$ as in Section 2; in particular $\beta = 2^{1064921} - 1$, and M_F is already shifted. The following definitions replace the earlier front end:

$$\begin{aligned} J &= 6j + 1, & q &= \beta J + 1, & W &= 12w + 4, \\ X &= qW, & Y &= qv, & E &= XY, \\ a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + 4a + 3. \end{aligned} \tag{9.1}$$

The transport parametrization is unchanged:

$$\begin{aligned} F &= (K_0 + W)C - z(q-1), & \alpha &= q - F - C, \\ R &= \big(q(q-F) - C\big)(q^2-1) + (M_C + qM_F)J, \\ k &= R + 1 + hE, & \eta &= c - Yk, & \zeta &= k - \eta. \end{aligned} \tag{9.2}$$

Put

$$\begin{aligned} i &= 2r + 1, & I &= ic^2, & D_f &= 1 + \Delta I^2, \\ Q &= \Delta^2 I^2, & \delta &= Q - 1, & V &= D_f(cT - R) - c, \\ C_y &= QV^2 - 1, & D_\tau &= 1 + XY^2(XY^2 + 1)k^2, \\ D_{\mathrm{m}} &= 1 + \Delta c^2, & b_0 &= X + ac. \end{aligned} \tag{9.3}$$

The rational notation $D_y = C_y/\delta$ is used only in the proof. Define

$$\begin{aligned} G &= (A_{\mathrm{cone}}F)(B_{\mathrm{cone}}C) - (q - A_{\mathrm{cone}}F - B_{\mathrm{cone}}C)^2, \\ A_{\mathrm{g}} &= v^2\big[G\eta\zeta - (t^2 - G - \eta\zeta + 1)^2\big] - 1 - p_2^2 - p_3(p_3+1). \end{aligned} \tag{9.4}$$

The multiplier v^2 in the guard is intentional. It will force $v \neq 0$, and its effect on the radical's magnitude will be controlled.

For the new carrier set

$$\begin{aligned} M &= 2q^2, & \Lambda &= 4H + 1, & U &= MH, \\ Z &= U(n+V) + M\Lambda b_0 + H(v - q^2), \\ A_0 &= M^2\Lambda^2 D_{\mathrm{m}}, & A_1 &= U^2 D_f, & A_2 &= U^2 D_\tau, \\ A_3 &= U^2 A_{\mathrm{g}}, & A_4^{\mathrm{num}} &= U^2 C_y. \end{aligned} \tag{9.5}$$

Here M is a carrier modulus, unrelated to the native mask numerals. The requested equation is the single ordinary polynomial

$$\boxed{P_{13}^{\mathrm{CRT}} = \delta^{16}\mathcal{F}_{32}\left(Z; A_0, A_1, A_2, A_3, \frac{A_4^{\mathrm{num}}}{\delta}\right) = 0.} \tag{9.6}$$

The denominator is cleared symbolically, exactly as in Section 3. No additional existential variable, variable exponent, numerical root extraction, or division operation is part of the polynomial.

9.2 Unconditional realness, including the zero-scale branch

At every integer assignment, q is even, $q \equiv 2 \pmod 3$, and $|q| \geq 16$. The factor $3w + 1$ never vanishes, so $|X| \geq 64$. The new $Y = qv$ may initially vanish; it must not be silently assumed nonzero.

If $v = 0$, then $Y = a = 0$, $\Delta = 3$, and $H = 3$. If $v \neq 0$, then $|Y| \geq 16$ and $|a| = |X+1||Y| \geq 63 \cdot 16$, so $\Delta > 0$ and $|H| > 3$. In both cases Δ is odd, $H \neq 0$, and

$$3 < \frac{\Lambda}{H} = 4 + \frac{1}{H} < 5. \tag{9.7}$$

In particular $\Lambda \neq 0$, and the carrier scales are nonzero.

The radicands D_{m} and D_f are at least one. With $u = XY^2 \in \mathbb{Z}$, the integer inequality $u(u+1) \geq 0$ gives $D_\tau = 1 + k^2 u(u+1) \geq 1$. For the auxiliary radicand, if $c = 0$, then $Q = 0, \delta = -1, C_y = -1$, and $D_y = 1$. If $c \neq 0$, then

$$Q = \Delta^2 i^2 c^4 > 1, \qquad D_f = 1 + \Delta i^2 c^4 > |c| + 1.$$

Writing $b = cT - R \in \mathbb{Z}$, either $b = 0$ and $|V| = |c| \geq 1$, or $b \neq 0$ and $|V| \geq D_f - |c| > 1$. Therefore

$$D_y = V^2 + \frac{V^2 - 1}{Q - 1} \geq 1.$$

We have proved, without using a guard, a Pell equation, or native decoding, that $\delta \neq 0$ and all four non-guard radicands are positive on every integer assignment.

At a zero of (9.6), one of the conjugate linear factors vanishes over $\mathbb{C}$. If $A_{\mathrm{g}} < 0$, its radical would be the only nonzero imaginary summand in a factor with a real center and four real summands. Thus $A_{\mathrm{g}} \geq 0$. It follows that

$$v^2 \operatorname{Pair}(G, \eta\zeta, t) = A_{\mathrm{g}} + 1 + p_2^2 + p_3(p_3 + 1) > 0. \tag{9.8}$$

Consequently $v \neq 0$ and $\operatorname{Pair}(G, \eta\zeta, t) > 0$. The joint-positivity lemma gives $G > 0$ and $\eta\zeta > 0$. The latter excludes $c = 0$: that case would give $\eta\zeta = -Y(Y+1)k^2 \leq 0$. Hence at a zero we have $c \neq 0, Q > 1, \delta > 0$.

This proves the required hypotheses in their proper order. In particular, $v = 0$ is excluded because its guard radicand is strictly negative; it was not excluded when proving unconditional realness.

9.3 Separation with only $|Y| \geq |q|$

Set

$$x = |X|, \qquad m_Y = |Y|, \qquad K = |k|, \qquad C_0 = |c|.$$

The recovered strict guards imply

$$x \geq 64, \quad m_Y \geq |q| \geq 16, \quad K \geq 2, \qquad K(m_Y - 1) < C_0 < K(m_Y + 1). \tag{9.9}$$

Indeed $\eta/k \in (0, 1)$ and $c = k(Y + \eta/k)$. In particular $C_0 > 30$. Let

$$d_+ = \sqrt{D_{\mathrm{m}}}, \quad f_+ = \sqrt{D_f}, \quad \tau_+ = \sqrt{D_\tau}, \quad y_+ = \sqrt{D_y}, \quad g_+ = \sqrt{A_{\mathrm{g}}}, \quad \lambda = \Lambda/H.$$

The guard root g_+ may be zero; the others are positive.

As before, the determinant inequality $G > 0$ proves $G < q^2$: with $U_c = A_{\mathrm{cone}}F$, $W_c = B_{\mathrm{cone}}C$, its two factors have the same sign and $|U_c| + |W_c| < 2|q|$. Since $\eta\zeta \leq k^2/4$, the scaled guard gives

$$g_+ < \frac{|v||q|K}{2} = \frac{Km_Y}{2} < \frac{8C_0}{15}. \tag{9.10}$$

The last bound uses $m_Y \geq 16$ and $C_0 > K(m_Y - 1)$.

The first norm gives

$$K(xm_Y^2-1)<\tau_+<K(xm_Y^2+1),\qquad \tau_+>2C_0. \tag{9.11}$$

The main root has a useful lower comparison with τ_+:

$$d_+>\big((x-1)m_Y-3\big)K(m_Y-1)>\frac{3}{4}\tau_+.$$

For the second inequality, expansion gives

$$\begin{aligned}&4\big((x-1)m_Y-3\big)(m_Y-1)-3(xm_Y^2+1)\\&\quad=m_Y^2(x-4)-4m_Y(x+2)+9>0\end{aligned}$$

for $x\ge 64, m_Y\ge 16$. Combining this with (9.7) deliberately places the scaled main root above the first root:

$$\lambda d_+>2\tau_+. \tag{9.12}$$

Also

$$f_+^2>\Delta C_0^4,\qquad d_+^2=\Delta C_0^2+1<2\Delta C_0^2,$$

so

$$f_+>\frac{C_0}{2}d_+>15d_+>2\lambda d_+. \tag{9.13}$$

The estimates $f_+>3C_0^2$ and $Q>f_+^2$ hold here too, since $|a|$ is large and $i\neq 0$. Thus the same two auxiliary cases remain valid:

$$\begin{cases}C_0<y_+<C_0+1, & cT-R=0,\\ y_+>2f_+, & cT-R\neq 0.\end{cases} \tag{9.14}$$

In the first case, $y_++g_+<(23/15)C_0+1<2C_0<\tau_+$. Equations (9.12) and (9.13) give strict superincreasing order

$$f_+,\quad \lambda d_+,\quad \tau_+,\quad y_+,\quad g_+.$$

In the second case the order is

$$y_+,\quad f_+,\quad \lambda d_+,\quad \tau_+,\quad g_+.$$

Zero g_+ is omitted. Each entry exceeds the sum of all entries to its right, not merely the next entry.

Dividing a vanishing factor by U expresses a rational number as a signed sum of precisely these magnitudes. Lemma 6.1 makes all five roots rational. The roots d, f, τ, p_1 of the integer radicands $D_{\mathrm{m}}, D_f, D_\tau, A_{\mathrm{g}}$ are integral, and Lemma 6.2 makes the rational auxiliary root y integral. The factor's signs can be absorbed into these integer roots.

9.4 Two coprime congruences, with no hidden scale promise

Because $Y=qv$,

$$H=4(X+1)qv+3\equiv 3 \pmod{q}.$$

The fixed odd exponent gives $\beta\equiv 1 \pmod 3$, and $J=6j+1$, so $q\equiv 2 \pmod 3$. Since H is odd,

$$\gcd(H,M)=\gcd(H,2q^2)=1,\qquad \Lambda\equiv 1 \pmod{H}. \tag{9.15}$$

These are universal arithmetic facts about the displayed formulas. They are not assertions about unspecified canonical witnesses.

The signed integer roots satisfy

$$U(n+V) + M\Lambda b_0 + H(v-q^2) = M\Lambda d + U(f+\tau+p_1+y). \tag{9.16}$$

Modulo M, this gives $H(v-q^2) \equiv 0$, so (9.15) yields

$$v = q^2(2v_{\text{old}}+1), \qquad v_{\text{old}} = \frac{v-q^2}{2q^2} \in \mathbb{Z}. \tag{9.17}$$

Consequently $Y = qv = (2v_{\text{old}}+1)q^3$, exactly the original source scale, including its odd quotient. Modulo H, (9.16) gives $M\Lambda(d-b_0) \equiv 0$. Both multipliers are units modulo H, so

$$\gamma = \frac{d-b_0}{H} \in \mathbb{Z}. \tag{9.18}$$

This restores the main Pell root and its required congruence.

The normalized strong and auxiliary witnesses are restored by (6.4)–(6.5), unchanged. Since $\mathrm{Pair}(G, \eta\zeta, t) > 0$, the universal ternary Pos representation supplies the old private guard variables. They need not equal the new ones: the latter represent $v^2\,\mathrm{Pair}(G, \eta\zeta, t)$. This distinction is necessary for soundness. The transport equation already holds identically by (9.2).

Theorem 9.1 (Coprime-carrier reduction)**.** *The polynomial P_{13}^{CRT} has an integer zero if and only if the normalized integer system $\mathcal{K}$ has an integer solution. Consequently, with the same inherited ordinary calibration,*

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in \mathbb{Z}^{13}\ P_{13}^{\mathrm{CRT}}(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

The polynomial has exact total degree 1088*.*

Proof. The preceding argument restores a solution of $\mathcal{K}$ from a polynomial zero. Conversely, start with a solution of $\mathcal{K}$, whose old scale is $Y = (2v_{\text{old}}+1)q^3$, and put

$$v = Y/q = q^2(2v_{\text{old}}+1) \neq 0.$$

All native quantities remain unchanged. Retain its t, for which $\mathrm{Pair}(G, \eta\zeta, t) > 0$, and regenerate p_1, p_2, p_3 as a Pos representation of the positive integer $v^2\,\mathrm{Pair}(G, \eta\zeta, t)$. Then the new $A_g = p_1^2$. Finally choose

$$n = \Lambda\gamma + f + \tau + p_1 + y - V - \frac{v-q^2}{M} \in \mathbb{Z}. \tag{9.19}$$

Every quantity on the right is fixed before n, and substitution gives (9.16). One norm factor therefore vanishes. This is the exact sign convention for the displayed centered carrier. The calibration conclusion composes this new algebraic equivalence with the inherited ordinary proof for $\mathcal{K}$. Exact degree is proved below. □

For a canonical refutation witness the same construction uses its actual Y; no extra divisibility or freedom of choice is asserted. The earlier odd strong normalization is also unchanged: with $A = a+2$ even and R, c odd, the recurrence gives $\psi_A(Rc)$ odd, and division by the odd c^2 gives odd i. This parity fact follows from the recurrence, not from only the first term of a binomial congruence.

9.5 Exact degree and a coefficient certificate

The lower-degree scale changes the relevant degrees to

register	a	Δ	D_f	Q	R	V	C_y	A_g
degree	4	8	14	22	5	19	60	30

We also have $\deg D_\tau = 22$, $\deg U = 6$, and $\deg Z = 25$. In the biquadratic evaluator of Section 3, take

$$d = M^2\Lambda^2 D_{\mathrm{m}}, \quad e = U^2 A_{\mathrm{g}}, \quad A = U^2 D_f, \quad B = U^2 D_\tau, \quad C_{\mathrm{n}} = U^2 C_y.$$

Their respective degrees are $22, 42, 26, 34, 72$. Assign formal weights 11 and 21 to the roots of d and e. Writing

$$s = Un + M\Lambda b_0 + H(v - q^2), \qquad Z = s + UV,$$

the scalar coefficient of the first norm register has the exact cut

$$J_0 = \delta(s^2 + 2UVs + d + e - A - B) - U^2(V^2 - 1). \tag{9.20}$$

Here $\deg s = 11$. The four weighted component degrees are

$$64, \quad 58, \quad 68, \quad 54.$$

The unique highest component is $2\delta Z\sqrt{e}$, of weight 68. The scalar subtraction in the next squaring has degree at most $128 < 136$, and the last tail subtraction has weight at most $248 < 272$. Therefore the cleared three-root norm has unique leading scalar

$$16 Q_0^4 Z_0^4 e_0^2$$

of degree 272. The two quadratic norm operations give exact degree 1088 and top form $2^{16} Q_0^{16} Z_0^{16} e_0^8$.

For a numerical-slice certificate, define

$$\begin{aligned}
&q_0 = 6\beta j, && W_0 = 12w, && Y_0 = q_0 v, && X_0 = q_0 W_0,\\
&a_0 = X_0 Y_0, && i_0 = 2r, && F_0 = W_0 C - q_0 z, && R_0 = -q_0^3 F_0,\\
&k_0 = R_0 + h a_0, && \eta_0 = -Y_0 k_0, && Q_0 = a_0^4 i_0^2 c^4,\\
&V_0 = -a_0^2 i_0^2 c^4 R_0, && U_0 = 8 q_0^2 a_0, && Z_0 = U_0 V_0,\\
&e_0 = -U_0^2 v^2 \eta_0^4.
\end{aligned}$$

Direct substitution gives the nonzero leading polynomial

$$\left(P_{13}^{\mathrm{CRT}}\right)_{1088} = 2^{112} q_0^{64} a_0^{128} i_0^{64} c^{128} R_0^{16} v^{16} \eta_0^{32}. \tag{9.21}$$

In particular,

$$\left[j^{480} w^{160} v^{208} h^{32} r^{64} c^{128} z^{16}\right] P_{13}^{\mathrm{CRT}} = 2^{976} 3^{640} \beta^{480} > 0. \tag{9.22}$$

The exponent h^{32} uniquely selects the $h a_0$ term in k_0 throughout η_0^{32}; z^{16} then uniquely selects the $q_0^4 z$ term throughout R_0^{16}. All remaining factors are monomials. The exponent sum is 1088, so this proves exact degree on the actual fixed coefficient slice, independently of possible relations among the other five numerals.

The gain comes from a complete carrier construction with a proved coprimality identity and explicit two-way witness maps. The inherited calibration and its stated formalization boundary remain exactly those identified in the earlier sections.

10 Four-root and three-root companions

Define $\mathcal{F}_{16}$ and $\mathcal{F}_8$ by the corresponding products over four and three signed radicals. Retain the base registers (2.1)–(2.4), and write

$$J_g = A_{\mathrm{g}} - p_1^2, \qquad \ell = XY^2 k, \qquad F_{\mathrm{s}} = (a+2)I.$$

A useful finalizer is

$$\mathcal{P} = N + S + NS, \qquad S \text{ a sum of integer squares.}$$

For integer inputs, $\mathcal{P} = 0$ is equivalent to $S = N = 0$: $(1+S)(1+N) = 1$ and $1+S$ is a positive integer.

10.1 Fourteen integers, degree 696

Add p_1 to the thirteen base coordinates and set

$$\begin{aligned} Z_{14} &= H(n + 2V + 2\ell + k + 2F_{\mathrm{s}}) + 2b_0, \\ N_{14} &= \delta^8 \mathcal{F}_{16}\left(Z_{14}; 4D_{\mathrm{m}}, 4H^2 D_f, 4H^2 D_\tau, \frac{4H^2 C_y}{\delta} \right), \\ S_{14} &= J_g^2, \qquad P_{14} = N_{14} + S_{14} + N_{14} S_{14}. \end{aligned} \tag{10.1}$$

The literal polynomial has 150 operations. At a zero, the guard and four-root separation give integral d, f, τ, y. A signed factor yields $Z_{14} = 2d + 2H(f + \tau + y)$. Hence $H \mid 2(d - b_0)$, and oddness of H gives $\gamma = (d - b_0)/H \in \mathbb{Z}$. Conversely, choose

$$n = 2\gamma + 2f + 2\tau + 2y - 2V - 2\ell - k - 2F_{\mathrm{s}}.$$

This is an existence-preserving half-centering, with no parity assumption on the independent coordinate n.

Here is a precise degree argument. Replace all independent variables by a common indeterminate multiple and work at infinity in the formal Laurent-series field. The three centered radicals have centers and corrections

center	radicand minus center squared	degrees: center, correction
$2HF_{\mathrm{s}}$	$4H^2(1 - I^2)$	$15, 18$
$H(2\ell + k)$	$H^2(4 - k^2)$	$23, 26$
$2HV$	$4H^2(V^2 - 1)/\delta$	$29, 28.$

Their root deviations have degrees at most $3, 3, -1$. Of the sixteen conjugate factors, eight first fail to cancel the last center, four first fail to cancel the middle center, and two first fail to cancel the first center. Their degrees are $29, 23, 15$, respectively. The final main-root pair has degree 14, with leading form $H_0^2 n(n + c)$, where $H_0 = 4a_0$. Clearing δ^8 adds 240. Therefore

$$\deg N_{14} = 8(29) + 4(23) + 2(15) + 14 + 240 = 608.$$

Its highest form is

$$(N_{14})_{608} = 2^{60} a_0^{66} i_0^{34} c^{68} R_0^8 \ell_0^4 n(n + c), \qquad \ell_0 = h X_0^2 Y_0^3.$$

The guard square has degree 88, led by η_0^8. Consequently $\deg P_{14} = 696$, certified, for example, by

$$[j^{396} w^{82} v^{94} h^{12} r^{34} c^{68} z^8 n^2] P_{14} = 2^{748} 3^{478} \beta^{396} > 0.$$

The unshifted fourteen-variable companion replaces the center and radicands by

$$Hn + b_0; \quad D_{\mathrm{m}}, H^2 D_f, H^2 D_\tau, H^2 C_y/\delta.$$

Its cleared norm has degree 704, led by $(H^2 C_y)_{88}^8$; the same finalizer has exact degree 792 and 137 operations. The four-root argument recovers the main residue directly.

10.2 Fifteen integers, degree 344

Add γ, p_1 to the thirteen base coordinates and use

$$\begin{aligned} Z_{15} &= n + 2V + 2\ell + k + 2F_{\mathrm{s}}, \\ N_{15} &= \delta^4 \mathcal{F}_8(Z_{15}; 4D_\tau, 4D_f, 4C_y/\delta), \\ S_{15} &= ((b_0 + H\gamma)^2 - D_{\mathrm{m}})^2 + J_g^2, \\ P_{15} &= N_{15} + S_{15} + N_{15}S_{15}. \end{aligned} \tag{10.2}$$

There are 131 operations. The two explicit residuals provide the main and guard rows. Three-root separation supplies integral f, τ, y, and conversely

$$n = 2(f + \tau + y) - 2V - 2\ell - k - 2F_{\mathrm{s}}$$

makes a norm factor vanish.

The Laurent centers $2V, 2\ell + k, 2F_{\mathrm{s}}$ have degrees $23, 17, 9$. Their deviations have degrees at most $-7, -3, -3$, so the final fully canceled factor has leading term n, degree one. The eight-factor norm therefore has degree

$$4(23) + 2(17) + 9 + 1 + 4(30) = 256.$$

Its leading form is $2^{14} Q_0^4 V_0^4 \ell_0^2 (F_{\mathrm{s}})_0 n \neq 0$. The guard square again has degree 88, giving exact final degree $256 + 88 = 344$. These degree computations are formal polynomial calculations; they do not use numerical asymptotics of integer solutions.

11 Four quotient-fusion companions

The archive also gives less radical-intensive equations with variable/degree pairs

$$(20, 24), \quad (19, 28), \quad (18, 32), \quad (17, 88).$$

They use the same six coefficients. Their full literal residual lists and independent certificates are in `companions/quotient_fusion/PROOF.md` and its JSON files.

For completeness, the twenty-variable presentation is specified here. Its independent coordinates are

$$j_3, F, \alpha, z, f, h, r, v, w_3, \tau, \eta, y, \gamma, T, a, c, t, p_1, p_2, p_3.$$

Use $J = 3j_3 + 1$, $q = \beta J + 1$, $W = 4(3w_3 + 1)$, $X = qW$, $Y = (2v + 1)q^3$, $C = q - F - \alpha$, and

$$\begin{aligned} E &= a - Y, & H &= 4a + 3, & \Delta &= a^2 + H, \\ R &= [q(q - F) - C](q^2 - 1) + (M_C + qM_F)J, & & & k &= R + 1 + hE. \end{aligned}$$

Set $I = (2r + 1)c^2$, $Q = \Delta(f^2 - 1)$, $B = f^2(cT - R)$, $\zeta = k - \eta$, and retain the determinant G of (2.4). Sum the squares of these eight residuals:

$$\begin{aligned} & a - (X + 1)Y, && c - \eta - Yk, \\ & \tau^2 + (2\tau - k)E(c - \eta) - 1, && (X + ac + \gamma H)^2 - \Delta c^2 - 1, \\ & (K_0 + W)C - F - z(q - 1), && I^2 - Q, \\ & (B + y)^2 - Q(c + y)(2B + y - c) - 1, \\ & G\eta\zeta - (t^2 - G - \eta\zeta + 1)^2 - \mathrm{Pos}(p_1, p_2, p_3). \end{aligned}$$

This polynomial has degree 24 and 120 operations. It centers the old first and auxiliary roots by

$$\tau_{\mathrm{old}} = E(c-\eta)+\tau, \qquad y_{\mathrm{old}} = B+y.$$

The auxiliary identity is $y_{\mathrm{old}}^2 - (B-c)^2 = (c+y)(2B+y-c)$.

Replacing F, α, z by C, z through the monic transport parameterization gives nineteen variables, degree 28, and 116 operations. Next substitute $a = (X+1)Y$, $E = XY$, remove the scale row, and use I^2 instead of Q in the auxiliary row. The strong residual makes this row replacement reversible on zeros. This gives eighteen variables, degree 32, and 112 operations. Finally substitute $\eta = c - Yk$ and delete its row, obtaining seventeen variables, degree 88, and 110 operations.

Soundness of quotient fusion must derive $c \mid f^2 - 1$ before restoring the removed quotient. The inherited main classification and ordinary strong-rank theorem give $c = \pm\psi_A(p)$, $|f| = \chi_A(m)$, $p \mid m$, hence $c^2 \mid f^2 - 1$. Only then define

$$o = f(cT - R), \qquad j_{\mathrm{old}} = f^2T - 1 - R(f^2-1)/c.$$

These are integers with $fo - c = B - c = cj_{\mathrm{old}} - R$. Completeness uses the squared-modulus canonical auxiliary choice already described, with the ordinary Δ-scaled odd strong multiplier. This avoids the invalid shortcut of canceling Δ from the ordinary strong equation.

The exact degree witnesses include

$$[f^8F^4j_3^{12}]P_{20} = (3\beta)^{12}, \qquad [f^8z^4j_3^{16}]P_{19} = (3\beta)^{16},$$

and the highest guard forms certify the eighteen- and seventeen-variable endpoints. The independent sparse verifier retains symbolic fixed ports until specializing them to the authentic tuple.

12 Reproduction and evidence

From the archive root, with Python 3.10 or later, run:

```
python3 build_equations.py
python3 verification/verify_release.py
python3 -O verification/verify_release.py
pdflatex -interaction=nonstopmode -halt-on-error \
  -output-directory=report report/report.tex
pdflatex -interaction=nonstopmode -halt-on-error \
  -output-directory=report report/report.tex
```

The release verification uses explicit checks that remain active under `-O`. Its results bind the complete numeral payloads, literal coefficient recipes, declared variables, live arithmetic gates, exact norm identity, degree certificates, and companion equations. The editable TEX files and the PDF are both delivered.

The finite evidence and the unbounded proof have different roles. Exact coefficient dictionaries establish universal polynomial identities. A nonzero symbolic highest form establishes degree for the actual fixed slice. Hashes and reconstruction bind the coefficient data. Sample evaluations can detect transcription mistakes, but cannot prove that all zeros decode, that a compiler is complete, or that the consistency biconditional holds. Those assertions use the mathematical arguments and imported ordinary calibration identified above.

The independent audits were independently implemented or derived within the research collaboration. They are not a claim of external human peer review. No full expansion of the gigantic numerical polynomial is needed: its literal division-free circuit and fixed integer leaves specify the ordinary polynomial exactly.

13 What is new and what remains to improve

The principal new composition has five parts: monic transport coordinates, an odd normalized strong witness, squared-modulus auxiliary restoration, recovery of the main Pell residue through the norm center, and absorption of the guard through unconditional realness followed by automatic magnitude separation. The last step removes the guard's square witness without the enormous separating weights required by a generic square-condition combiner. The CRT companion further recovers the odd cubic scale and the main residue from one integral carrier; its weaker scale is safe because the scaled guard excludes the new zero-scale boundary first.

Several ingredients are inherited: the radical norm, joint-positivity lemma, determinant cone, adjacent-coefficient descent, Pell congruences, and native/refutation calibration. We claim the displayed new reductions and literal measured endpoints, not worldwide priority for their general ingredients. The proposal remains K215-based through its fixed coefficient tuple. The request did not require avoiding that source, and the structural simplifications do not depend on Jones's universal polynomial.

There is no established global minimum under all reasonable size measures. Fewer integer variables can demand much higher degree; smaller circuits can have much larger coefficients. The degree-1088 form and the 186-operation degree-1392 form express that tradeoff concretely. A useful next target is a square-extraction argument that merges another native coordinate without losing the guard's quantitative information. Any such attempt must preserve canonical completeness, not merely the forward decoding direction. A separate substantial task is to compose the literal native instance and this reduction in the repository's sealed object-ZFC calculus.

A Complete bounded coefficient recipe

The following is the full source-34 finite arithmetic recipe [9]. All ranges are bounded constants. The ordered 943 tile records $(x_{i0}, x_{i1}, x_{i2}, x_{i3})$, in west, north, east, south order, are supplied in `coefficients/recipe/wang943_tiles.json`. Their ordering is part of the definition: selector 0 is Start and selector 1 is Forbidden. The supplied `k215.tm` contains all 430 working actions from which the source builder reconstructs the whole ordered list. Its SHA-256 is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`.

Thus this finite table is delivered data, not an unspecified machine or a free coefficient parameter.

Put

$$b = 121, \quad L_0 = 8801, \quad d = bL_0 = 1064921, \quad \rho = 2^{121}, \quad B_0 = 2^{1064921}.$$

For $0 \le s < 4$, $0 \le j < 10$, define

$$e_s(j) = o_s + 2j, \qquad (o_0, o_1, o_2, o_3) = (1020, 1000, 980, 960),$$

and $w_j = 2^{10+2j}$ for $0 \le j < 44$. Let

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{e_s(j) : 0 \le s < 4,\ 0 \le j < 10\} \\ & \cup \{1063, 1064, 1095, 1096, 2195, 3296\}.\end{aligned}$$

It has 989 elements and maximum 3296. For each tile $0 \le i < 943$, set

$$c_i = 1 + \sum_{s=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_{is} + 512) w_{10s+j} + \sum_{j=40}^{43} w_j.$$

Here $\mathrm{bit}_j(m) = \lfloor m/2^j \rfloor \bmod 2$ is used only to define the finitely many fixed integer coefficients. It is not an operation on a quantified Diophantine variable. The other coefficients are

$$c_{e_s(j)} = w_{10s+j}, \quad (c_{1095}, c_{1096}, c_{2195}, c_{3296}) = (w_{40}, w_{41}, w_{42}, w_{43}), \quad c_{1063} = c_{1064} = 0.$$

Define

$$\begin{aligned}
\mu &= 1022 + \sum_{j=0}^{43} w_j, \\
Z_{\mathrm{pad}} &= (2^{119} - 1) \sum_{e=8792}^{8798} \rho^e + (2^{27} - 1)\rho^{8799}, \\
D_C &= \rho^{40} + \rho^{41} + \rho^{1100} + \rho^{2200} + \sum_{e \in \mathcal{E}} c_e(\rho^{4393-e} + \rho^{5495-e}), \\
D_R &= \rho, \\
M_C &= B_0 - 1 - \sum_{\substack{e \in \mathcal{E} \\ e \neq 1}} \rho^e - 2\rho^{1063}, \\
M_{F,\mathrm{native}} &= 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} (\rho^{1000+2j} + \rho^{1021+2j}) + \rho^{2195} + \rho^{3296} + Z_{\mathrm{pad}}.
\end{aligned}$$

The six actual fixed ports are precisely

$$\boxed{\begin{array}{c} \beta = B_0 - 1, \quad K_0 = D_C + B_0 D_R, \quad M_C, \quad M_F = M_{F,\mathrm{native}} + B_0 - 1, \\ A_{\mathrm{cone}} = 2^{1113}, \qquad B_{\mathrm{cone}} = 2^{666104}. \end{array}}$$

All displayed exponents are fixed integers. In particular, M_F already includes its shift; shifting it again would change the equation and invalidate the binding.

Fixed numeral	Bit length	Minimal unsigned bytes
β	1 064 921	133 116
K_0	1 065 043	133 131
M_C	1 064 921	133 116
M_F	1 064 922	133 116
A_{cone}	1 114	140
B_{cone}	666 105	83 264
Total		615 883

Table 2: Exact fixed payload sizes before compression. The archive stores deterministic gzip encodings of the minimal unsigned little-endian bytes and binds their raw hashes.

The source proof verifies the whole coefficient interface, not just these sizes. It establishes local tile decoding, carry-free geometry, whole-cell rotation, mask populations, positive canonical slack, and an explicit rank controller. In particular

$$\mathrm{pc}(M_C) + \mathrm{pc}(M_{F,\mathrm{native}}) = d, \quad M_C \equiv 2 \pmod 4, \quad M_{F,\mathrm{native}} \equiv 4 \pmod 8.$$

The controller uses $d = 11^2 \cdot 13 \cdot 677$, $\lambda = 2341$, and a proved Boolean subset coverage lemma; it does not assume that d is a power of five. Its construction supplies odd canonical exponents, odd cubic scale, and the parity promises used in Lemma 7.1. The complete inherited proof and literal builder are included, with their internal source pins and licenses preserved.

B Archive map and integration

Location	Contents
`report/`	This editable report, the CRT section, and the compiled PDF.
`build_equations.py`, `five_square_norm.py`	Literal equation generator and division-free universal norm evaluator.
`equations/`	The six norm circuits, declared variable lists, exact degree metadata, and operation ledgers.
`coefficients/`	All six fixed numeral payloads, binding manifest, complete bounded recipe, machine table, and ordered tile list.
`companions/quotient_fusion/`	The four quotient-fusion equations, source proof, generator, and independent verifier.
`verification/`	Integrated release audit, exact norm identity and degree checks, and machine-readable receipts.
`provenance/`	Inherited interfaces, calibration status, independent mathematical audit, and reviewed frontier.
`provenance/upstream_ordinary/`	Verbatim ordinary calibration proofs and exact external endpoint sources, each bound by commit, path, SHA-256, and byte length.
`licenses/`	Preserved repository, source, and machine notices.

The folder is intended for a new proposal directory in the ProveIt Diophantine-consistency collection. No existing proposal number is reserved. All runtime paths are relative to the delivered folder. Keep the whole six-numeral binding together when copying or adapting a circuit. The coefficients, source system, and formula are a single fixed instance.

References

[1] V. Reshetnikov, *ProveIt*, repository snapshot `9a33342a03d19cf5b061b2083c6b5d3f27f752ed`, 24 October 2026. https://github.com/VladimirReshetnikov/ProveIt.

[2] ProveIt, *Con-ZFC Diophantine forms*, research catalogue and frontier at the stated snapshot. The exact catalogue README is retained as `provenance/CON_ZFC_FORMS_README.md`. https://github.com/VladimirReshetnikov/ProveIt/tree/9a33342a03d19cf5b061b2083c6b5d3f27f752ed/Computability/DiophantineConsistency/con-zfc-diophantine-forms.

[3] ProveIt, *Incoming research packets*, including *Structural Reductions, Round 11* and *ZFC Diophantine Joint Guards*, 24 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/9a33342a03d19cf5b061b2083c6b5d3f27f752ed/docs/incoming.

[4] ProveIt, *The object-language ZFC obligation*, `Computability/DiophantineConsistency/OBJECT_ZFC.md`. Verbatim boundary document retained as `provenance/FORMALIZATION_BOUNDARY.md` in this archive.

[5] ProveIt, *Ordinary proof of the combined215 machine*; *Ordinary strong norms and signed witnesses for the blank computation*; and *Native arithmetic and the remaining calibration contract*. Exact copies of the ordinary proof texts, source paths, and hashes are in `provenance/upstream_ordinary/SOURCES.json`. https://github.com/VladimirReshetnikov/ProveIt/blob/9a33342a03d19cf5b061b2083c6b5d3f27f752ed/Computability/BusyBeaver/Research/ZFC/candidate-k215/proof.md.

[6] ProveIt incoming packet, *ZFC Diophantine Joint Guards*, 24 October 2026, `joint/PROOF.md` and `guards/PROOF.md`. Relevant ordinary interfaces are retained in `provenance/JOINT_GUARD_AND_PROJECTIONS.md` and `provenance/RESIDUE_AND_DETERMINANT.md`.

[7] ProveIt, source 70 *Residue orientation* and source 71 *Determinant guards*; signed bootstrap and exact canonical coefficient composition. Pinned files retained in the archive's provenance directory.

[8] ProveIt, source 51, *Squared-modulus auxiliary reconstruction*, `51-sparse-arith/short/SQUARED_MODULUS_PROOF.md`; and source 61, positive auxiliary reconstruction lemma. The squared-modulus proof is retained as `provenance/SQUARED_MODULUS_PROOF.md`.

[9] ProveIt, source 34, *A mixed-radix, parity-separated, unrounded coefficient tuple*, `34-integer14/coefficients/COEFFICIENT_PROOF.md`. Complete recipe, materialized coefficients, and verbatim proof in this archive.

[10] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, revised 2 September 2026. https://arxiv.org/abs/2609.01594.

[11] Z.-W. Sun, *Further results on Hilbert's Tenth Problem*, Science China Mathematics **64** (2021), 281–306. https://arxiv.org/abs/1704.03504.

[12] J. P. Jones, *Universal Diophantine equation*, Journal of Symbolic Logic **47** (1982), 549–571. https://doi.org/10.2307/2273588.